WASP-36b

R-0860

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_230111` · created 2026-07-28T21:49:06+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459955.9337 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.047
Transit depth (Rp/Rs)^2	2.52 +/- 1.5 [%]
Orbital Inclination (inc)	87.84 +/- 2.94
Airmass coefficient 1 (a1)	1.36 +/- 0.21
Airmass coefficient 2 (a2)	-0.16 +/- 0.074
Scatter in the residuals of the lightcurve fit is	16.54 %
Best Comparison Star	#1 - [211, 192]
Optimal Aperture	2.53
Optimal Annulus	11.33
Transit Duration (day)	0.0925 +/- 0.0087

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.047 vs 0.1368 (deviation 0.47σ)
Duration fit vs published	0.0925 d vs 0.0758 d (deviation 1.92σ)
Mid-time O–C	-11.9 min
Depth SNR	3.4
Residual scatter	16.54 %

Light curve

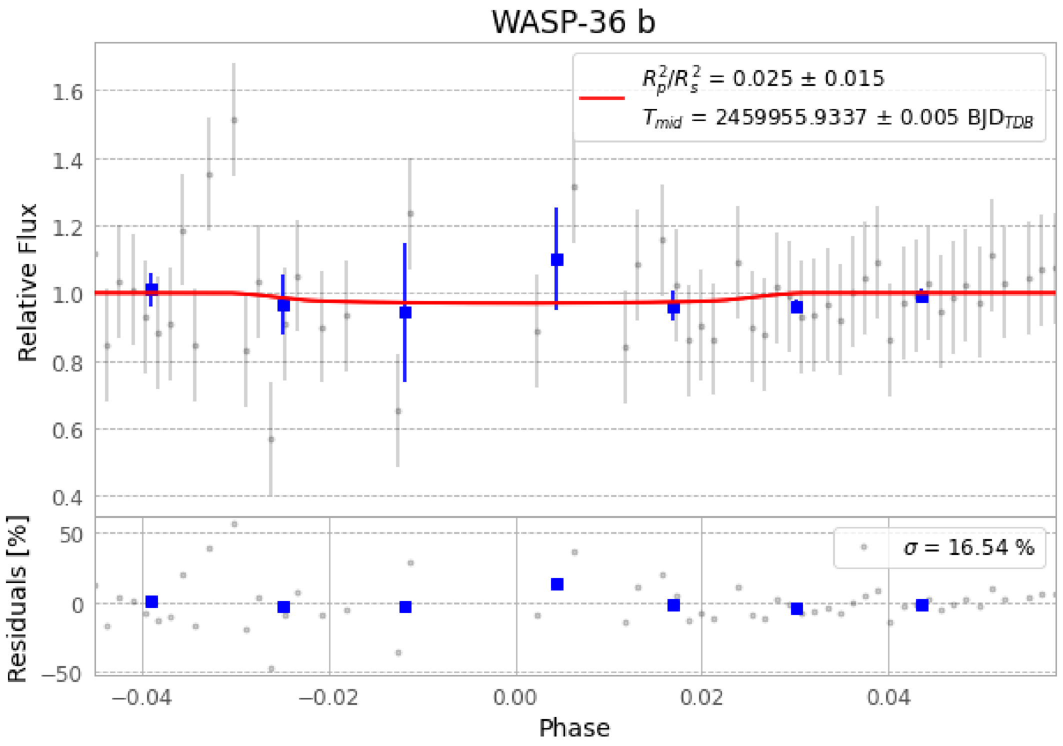

Figure 1 — FinalLightCurve_WASP-36 b_2023-01-11.png (SHA-256 1e10f0947cb5c8c7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 183], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c03c5e3230ae310c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

76 FITS frames (50 MB), `WASP-36230111083618.FITS` → `WASP-36230111122415.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-230111.log (12879ef05cb1…), FinalLightCurve_WASP-36 b_2023-01-11.png (1e10f0947cb5…), FinalLightCurve_WASP-36 b_2023-01-11.csv (91c8f1d874e1…)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 21:49Z.